Course: B.Tech | Stream: Computer Science | Subject: Data Structure & Algorithm | Sem: 3rd | Paper Code: PCC-CS301

Unit 2

Topics: ADT Stack & Expression Evaluation; ADT Queue Types (Simple, Circular, Priority) | Total Marks: 20 | Total Time: 45 minutes

Q1. Define ADT Stack and mention its two primary operations along with their function. **[2 Marks, L1]**

Q2. Explain the difference between a Simple Queue and a Circular Queue, with examples. **[3 Marks, L2]**

Q3. Write an algorithm to convert an infix expression to postfix using a stack, and trace it for the expression (A + B) * (C – D). **[5 Marks, L3]**

Q4. For a queue implemented using a simple (linear) array, analyze and compare the time complexity and space utilization of enqueue/dequeue operations versus a circular queue implementation. Explain why they differ. **[5 Marks, L4]**

Q5. A print-job scheduler must always execute high-priority jobs before regular ones, regardless of arrival order. Evaluate whether a Simple Queue is a suitable choice for this, and justify a better alternative if not. **[5 Marks, L5]**

--- End of Assignment ---